

REC Guardian - Frontend & UX Architecture Guide

This document details the frontend architecture, institutional design system, user experience workflows, component hierarchy, and synchronization between the dual frontend implementations in **REC Guardian**.

1. Design System & Visual Theme: Leaves Green & White (Light Mode)

1.1 Aesthetic Inspiration: Team Presentation Deck

In alignment with the official team presentation deck for **Team: KHATRON KE KHILADI** ("*Renewable Energy Certificate Fraud Detection System*"), the frontend is built exclusively in **pure Light Mode** using a **Leaves Green and White** botanical design system.

LEAVES GREEN & WHITE LIGHT-MODE DESIGN SYSTEM			
Canvas Surface Pure White / Tint #ffffff / #f8faf7	Deep Forest Green Primary Brand & H1 #143d2b / #0c2419	Leaf Emerald Cards & Badges #15803d / #16a34a	Lime Accent Trim Sprout Highlights #84cc16 / #65a30d

1.2 Core Color Tokens & Semantics

Token Name	Hex Code	UI Semantic & Application
Porcelain Mint Canvas	#f8faf7	Main page background; bright, clean, organic, and easy on the eyes.
Card & Surface Pure White	#ffffff	Elevated cards, dialog surfaces, table containers.
Border Sage Muted	#e2ece5 / #d0e0d5	Structural borders, subtle dividers, inactive pill borders.
Deep Forest Green	#143d2b / #0c2419	Hero banner gradient, high-contrast headings, primary text.
Vibrant Leaf Green	#15803d / #166534	Primary actions, verified badges, active navigation indicators.
Sprout Lime Trim	#84cc16 / #a3e635	Mottos, pill borders (" <i>DETECT. EXPLAIN. INVESTIGATE. PRESERVE EVIDENCE.</i> "), twin-leaf logo accent.
Forensic Warning Amber	#b45309 (on #fffbeb)	Medium risk claims, pending review notices, non-critical warnings.
Forensic Alert Crimson	#b91c1c (on #fef2f2)	High/Critical fraud holds, wash trading cycles, ledger tampering alerts.

2. shadcn/ui Component Architecture

The frontend components in `frontend/src/components/ui/` adhere strictly to the **shadcn/ui** design patterns:

Component	File Path	Description & Variants
Button	<code>frontend/src/components/ui/button.tsx</code>	Variants: <code>default</code> (Leaf 700), <code>outline</code> (Sage 200), <code>destructive</code> (Red 600), <code>ghost</code> , <code>link</code> , <code>sprout</code> (Lime).
Card	<code>frontend/src/components/ui/card.tsx</code>	<code>Card</code> , <code>CardHeader</code> , <code>CardTitle</code> , <code>CardDescription</code> , <code>CardContent</code> , <code>CardFooter</code> with <code>#e2ece5</code> borders and soft shadows.
Badge	<code>frontend/src/components/ui/badge.tsx</code>	Variants: <code>default</code> , <code>secondary</code> , <code>destructive</code> , <code>outline</code> , <code>success</code> , <code>warning</code> , <code>lime</code> , <code>forest</code> .
Tabs	<code>frontend/src/components/ui/tabs.tsx</code>	Segmented control with light sage background and active white pill.
Dialog	<code>frontend/src/components/ui/dialog.tsx</code>	Floating accessible modal with backdrop blur (<code>bg-forest-950/40</code>) and smooth enter animations.
Table	<code>frontend/src/components/ui/table.tsx</code>	Horizontal scroll wrapper (<code>overflow-x-auto</code>) with mint header (<code>bg-sage-100/70</code>) and cell hover highlights.
Input / Label	<code>frontend/src/components/ui/input.tsx</code>	Crisp input with subtle leaf-green focus ring (<code>focus-visible:ring-leaf-600</code>).
Select	<code>frontend/src/components/ui/select.tsx</code>	Styled dropdown selector matching the botanical palette.
Progress	<code>frontend/src/components/ui/progress.tsx</code>	Rounded progress bar with smooth leaf-green fill transition.
Sheet	<code>frontend/src/components/ui/sheet.tsx</code>	Slide-out drawer for 100% mobile and tablet responsiveness.
Alert	<code>frontend/src/components/ui/alert.tsx</code>	Notification banners with <code>default</code> , <code>destructive</code> , <code>warning</code> , <code>success</code> , and <code>forest</code> variants.
Separator	<code>frontend/src/components/ui/separator.tsx</code>	Structural dividing rules.

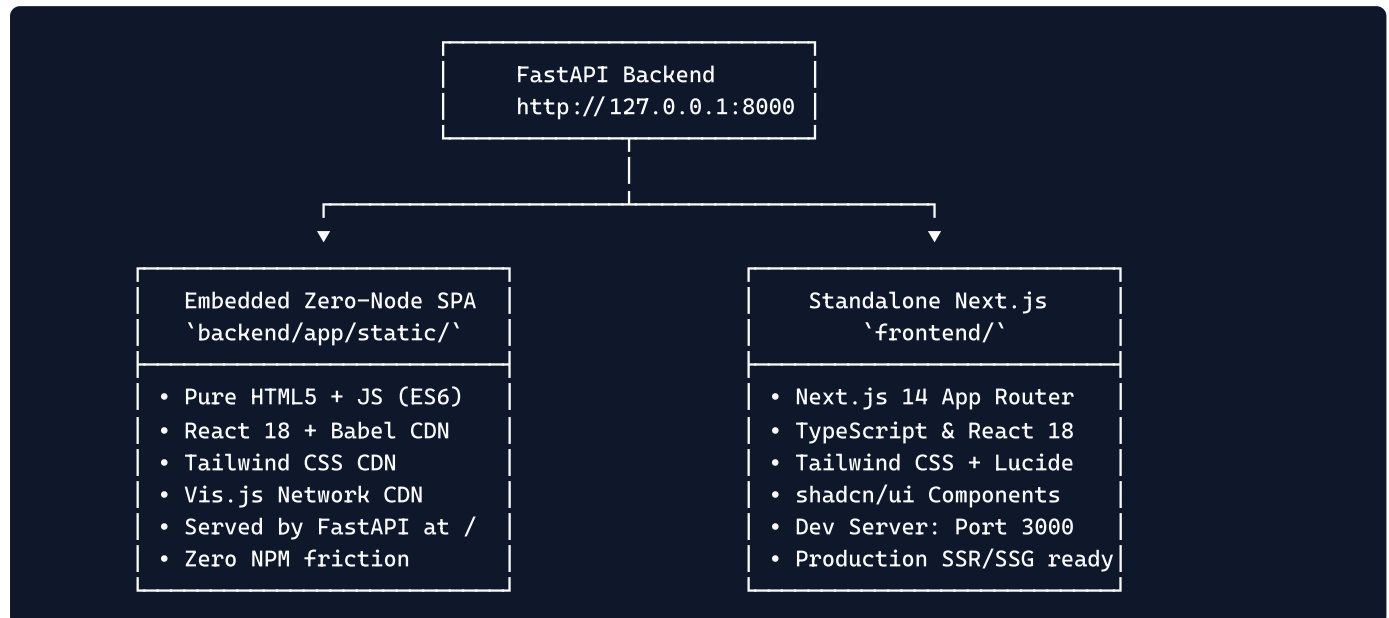
3. Responsive Layout & Mobile Experience

The interface is engineered for responsiveness across all screen dimensions:

- **Mobile (< 768px):** - Header displays twin-leaf logo, title, and a **hamburger menu button**. - Clicking hamburger opens the slide-out `<Sheet>` **Navigation Drawer** with direct access to all 7 modules. - KPI metric cards automatically collapse to a single-column layout (`grid-cols-1`). - All data tables are wrapped in smooth horizontal scroll containers (`overflow-x-auto`) with minimum table widths to prevent awkward text wrapping. - Floating dialogs fit comfortably on mobile screens with scrollable content bodies (`max-h-[90vh]`).
- **Tablet (768px - 1024px):** - 2-column grid for metric cards (`grid-cols-`

2). - 2-column layout for the 6-stage Lineage Explorer cards. - **Desktop (> 1024px)**: - Full horizontal tab strip with icon badges. - 4-column metric cards (`lg:grid-cols-4`). - 3-column / 6-stage lineage provenance grid with vertical chronological timeline. - Full-width Vis.js interactive physics canvas.

4. Dual Frontend Implementations



Both implementations share identical visual styling, color tokens, and workflows, ensuring that evaluators running `python backend/run.py` experience the exact same high-polish light-mode GovTech interface as frontend engineers running the Next.js application.

5. Running the Frontends

Option A: Embedded Live SPA (Instant, Zero Setup)

```
python backend/run.py
```

Open <http://127.0.0.1:8000> in any browser.

Option B: Next.js 14 Production App

```
cd frontend
npm install
npm run dev
```

Open <http://localhost:3000> with hot module reloading.